Hokeun Cha

hcha@cs.wisc.edu
https://chahk0129.github.io/

Research Interests

Database systems, storage systems, distributed systems

- Modern system architectures: time-series databases, disaggregated databases
- Core database techniques in transaction processing: indexing, concurrency control
- HW-SW co-design: RDMA, SmartNIC, CXL memory, byte-addressable storage

Education

University of Wisconsin-Madison — Madison, WI
Ph.D. in Computer Science — 2021 – 2027 *(expected)*

- Advisor: Prof. Xiangyao Yu

Sungkyunkwan University — South Korea
M.S. in Electrical and Computer Engineering — 2018 – 2020

- Thesis: Leveraging Byte-Addressability of Persistent Memory for B+-tree
- Advisor Prof. Beomseok Nam

B.S. in Computer Science and Engineering — 2012 – 2018
B.B.A. in Business Administration — (incl. Korean military service)

Work Experience

University of Wisconsin-Madison — Madison, WI
Research Assistant, Database Group — 2021 – Present

- Designed and implemented B^{link}*-hash*, a high-performance in-memory index that adapts to workload changes from heavy writes to intensive reads
- Developed *Lotus*, a two-sided RDMA transaction processing mechanism that minimizes network traffic and guarantees starvation freedom in disaggregated memory databases
- Designed *PoCo-Tree*, a lightweight disk-based index that minimizes index footprint via pointer coalescing while providing efficient query performance

Microsoft Research — Redmond, WA
Research Intern, Data Systems Group — Summer 2026

- Mentors: Badrish Chandramouli, Philip A. Bernstein, and Vasileios Zois
- Developed a high-performance distributed transactional system

Microsoft — Redmond, WA
Research Intern, Gray Systems Lab (GSL) — Summer 2025

- Mentors: Rathijit Sen and Matteo Interlandi
- Built a transaction processing system for disaggregated databases
- Designed a low-latency concurrency control mechanism for high-performance networks *(patent)*

Intel — Hillsboro, OR
Software Engineering Intern, Data Platforms Future Technology Team — Summer 2022

- Evaluated cost-performance analysis of storage devices in data-center system environments.
- Performed end-to-end infrastructure optimizations for database platforms

Sungkyunkwan University — South Korea
Researcher, Convergence Research Institute — 2020

- Built the prototype of *BallonStasher*, an RDMA-based disaggregated memory that improves resource utilization and system performance via memory sharing

Research Assistant, Data-Intensive Computing Lab — 2018 – 2020

- Developed B^3*-tree*, a write-optimized B+-tree for persistent memory that minimizes write traffic by employing logical representation of binary search tree in index nodes

Publications

[1] *Two-sided RDMA Striking Back for Disaggregated Memory Databases*,
Hokeun Cha, Aditya Akella, Xiangyao Yu,
arXiv preprint, Aug. 2026

[2] *Disaggregated Memory for File-backed Pages*,
Daegyu Han, Jaeyoon Nam, **Hokeun Cha**, Changdae Kim, Kwangwon Koh, Taehoon Kim, Sang-Hoon Kim, Beomseok Nam,
ACM Transactions on Storage (**TOS**), Nov. 2025

[3] *Towards Accelerating Data Intensive Application's Shuffle Process Using SmartNICs*,
Jiaxin Lin, Tao Ji, Xiangpeng Hao, **Hokeun Cha**, Yanfang Le, Xiangyao Yu, Aditya Akella,
Proceedings of ACM on Measurement and Analysis of Computing Systems (**SIGMETRICS**), Jun. 2023

[4] *B^{link}-hash: An Adaptive Hybrid Index for In-Memory Time-Series Databases*,
Hokeun Cha, Xiangpeng Hao, Tianzheng Wang, Huanchen Zhang, Aditya Akella, Xiangyao Yu,
Proceedings of **VLDB** Endowment, Feb. 2023

[5] *Pivotal B+ tree for Byte-Addressable Persistent Memory*,
Jonghyeon Yoo, **Hokeun Cha**, Wonbae Kim, Wook-Hee Kim, Sung-Soon Park, Beomseok Nam,
IEEE Access, Apr. 2022

[6] *B^3-tree: Byte-Addressable Binary B-Tree for Persistent Memory*,
Hokeun Cha, Moohyeon Nam, Kibeom Jin, Jiwon Seo, Beomseok Nam,
ACM Transactions on Storage (**TOS**), Jul. 2020

[7] *Write-Optimized Dynamic Hashing for Persistent Memory*,
Moohyeon Nam, **Hokeun Cha**, Young-ri Choi, Sam H. Noh, Beomseok Nam,
Proceedings of 17th USENIX Conference on File and Storage Technologies (**FAST**), Feb. 2019

Work-In-Progress

[1] PoCo-Tree: Pointer Coalesced Tree for Time-Series Databases
Hokeun Cha, Haojun Lioa, Jeff Tao, Xiangyao Yu

Talks

B^{link}-hash: An Adaptive Hybrid Index for In-Memory Time-Series Databases
- Database Group Seminar, Cornell University, Ithaca, NY *(virtual)* — Feb. 2024
- VLDB Conference, Vancouver, Canada, — Feb. 2023
- Database Affiliates Workshop, UW-Madison, Madison, WI — Oct. 2022

Write-Optimized Dynamic Hashing for Persistent Memory
- Korea Software Congress (KSC), Pyeongchang, South Korea — Dec. 2019

Teaching Experience

University of Wisconsin-Madison — Madison, WI
Teaching Assistant
- CS564: Database Management Systems — Spring 2021

Sungkyunkwan University — South Korea
Teaching Assistant
- SWE3021: Multi-core Computing — Fall 2019
- SWE3006: Programming Languages — Spring 2019

Honors

Computer Science Departmental Scholarship — 2021
University of Wisconsin-Madison, Madison, WI

Merit-based Scholarship for Graduate Students — 2018 – 2020
Sungkyunkwan University, Seoul, South Korea

Merit-based Scholarship for Undergraduate Students — 2013
Sungkyunkwan University, Seoul, South Korea

Service

Reviewing Activities
Journal of Data Science and Engineering — 2026
Journal of Cluster Computing — 2024
Journal of Supercomputing — 2023, 2024